Skill-Bridge Career Navigator

Karthik Mannem — mannemkarthik01@gmail.com

Demo Video: Watch the Full Demo on YouTube

Abstract

Skill Bridge is a modular career intelligence system designed to analyze structured and unstructured user input, retrieve relevant job requirements, compute demand-weighted skill gaps, and generate an actionable learning roadmap. The system balances AI-assisted extraction with deterministic scoring logic to ensure explainability, reliability, and reproducibility. This report describes the system architecture, hybrid extraction strategy, retrieval pipeline, gap analysis engine, and engineering tradeoffs.

1 Introduction

Early-career professionals often struggle to understand how their current skill set maps to real-world job requirements. Job postings are unstructured, inconsistent, and distributed across platforms, making it difficult to identify concrete next steps.

Skill Bridge addresses this problem by building an end-to-end system that:

- Parses synthetic resume and GitHub-style input,
- Retrieves relevant job requirements from a local dataset,
- Computes demand-weighted skill gaps,
- Generates a structured learning roadmap,
- Produces tailored interview questions.

The primary design goal was to implement a modular and explainable system rather than an opaque generative tool.

2 Technical Stack

The system was implemented using the following technologies:

- **Language:** Python 3.10+
- **Frontend Framework:** Streamlit
- **Embedding Model:** OpenAI text-embedding-3-small
- **Vector Index:** FAISS (CPU version)
- **Schema Validation:** Pydantic
- **Data Handling:** JSON / JSONL (synthetic datasets)
- **Testing Framework:** Pytest
- **Environment Management:** Python virtual environment + .env for secrets

3 System Architecture

The system follows a layered architecture consisting of input, intelligence, and output layers.

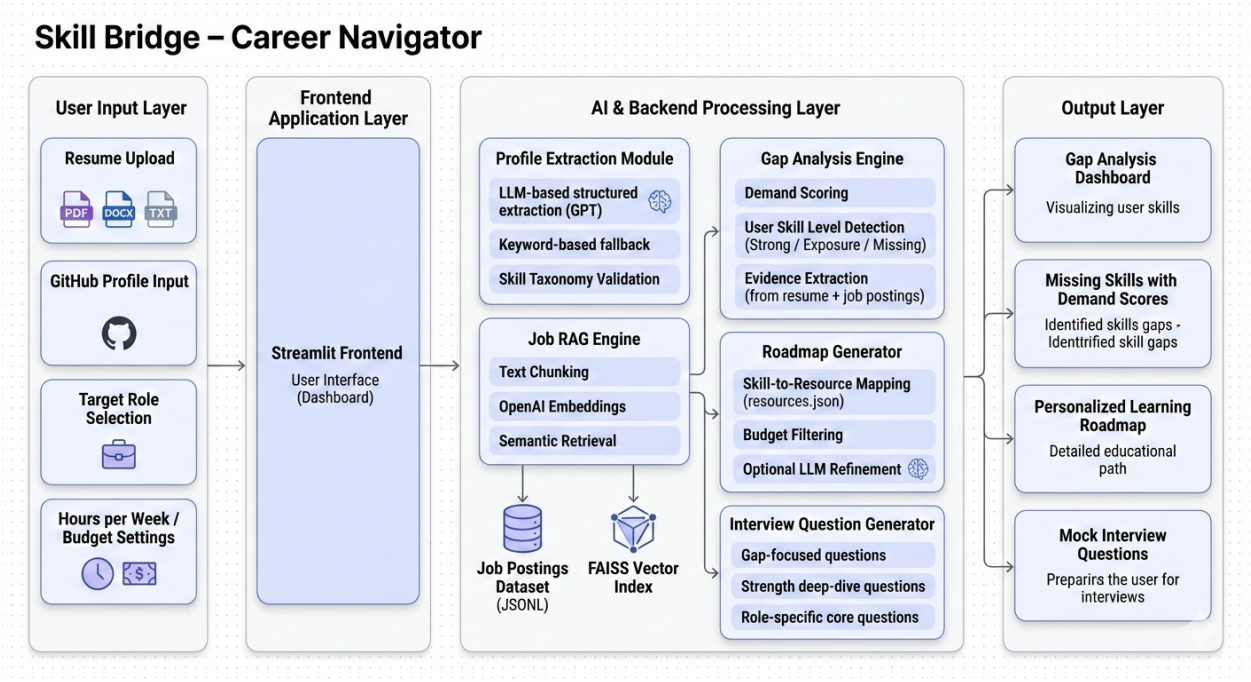

Figure 1: High-Level System Architecture

3.1 Input Layer

The system accepts:

- Resume files (PDF, DOCX, TXT),
- Optional GitHub profile metadata,
- Target role selection,
- Time and budget constraints.

Basic validation ensures correct file types and non-empty input.

3.2 Intelligence Layer

This layer contains five core modules:

1. Hybrid Profile Extraction
2. Job Requirement Retrieval
3. Demand-Weighted Gap Analysis
4. Roadmap Generator
5. Interview Question Generator

Each module operates independently, enabling extensibility and easier testing.

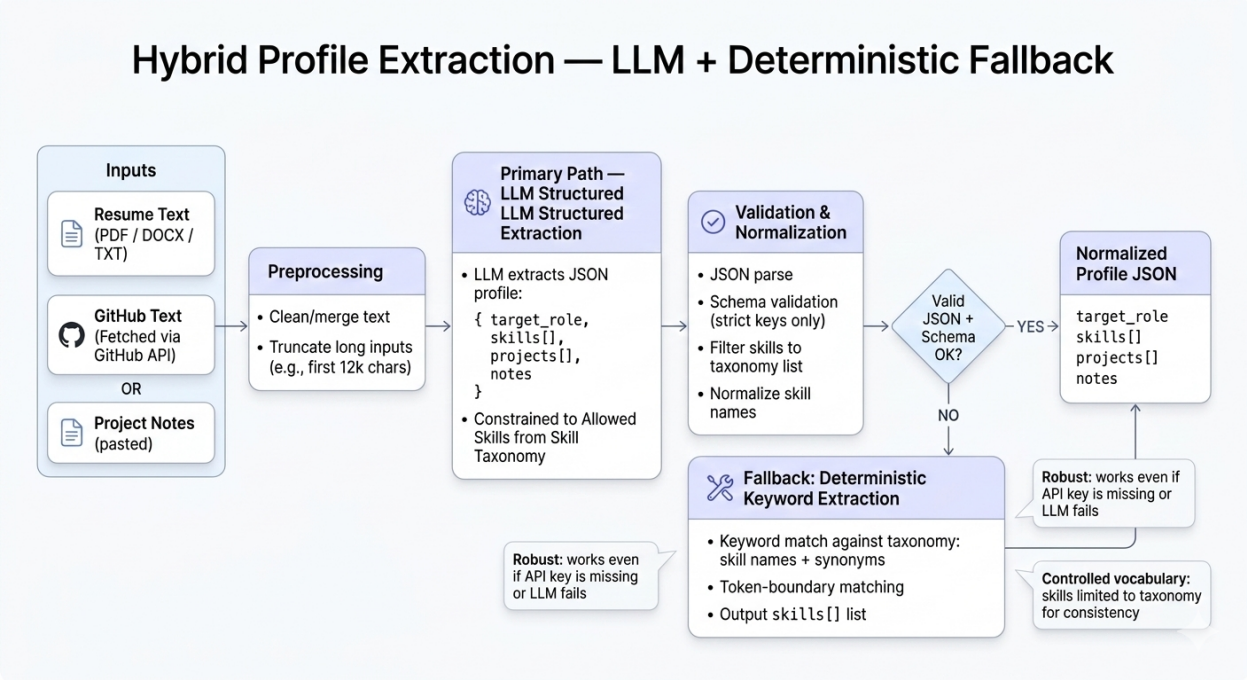

Figure 2: Hybrid Extraction Strategy

4 Hybrid Profile Extraction

Resume parsing is inherently noisy due to unstructured formatting. To address this, Skill Bridge implements a dual-path extraction strategy.

4.1 Primary Path: Structured AI Extraction

The system attempts to extract structured JSON containing:

- Target role,
- Skill list,
- Project summaries,
- Notes.

The output is validated against a strict schema to ensure correctness. Extracted skills are filtered against a predefined taxonomy to prevent hallucinated entries.

4.2 Fallback Path: Deterministic Keyword Extraction

If the AI response is malformed or unavailable, the system falls back to token-boundary keyword matching using skill synonyms defined in the taxonomy. This ensures consistent behavior even without API access.

5 Retrieval-Augmented Job Analysis

Job postings are stored locally in synthetic JSONL format. The retrieval pipeline consists of:

1. Text chunking with overlap,

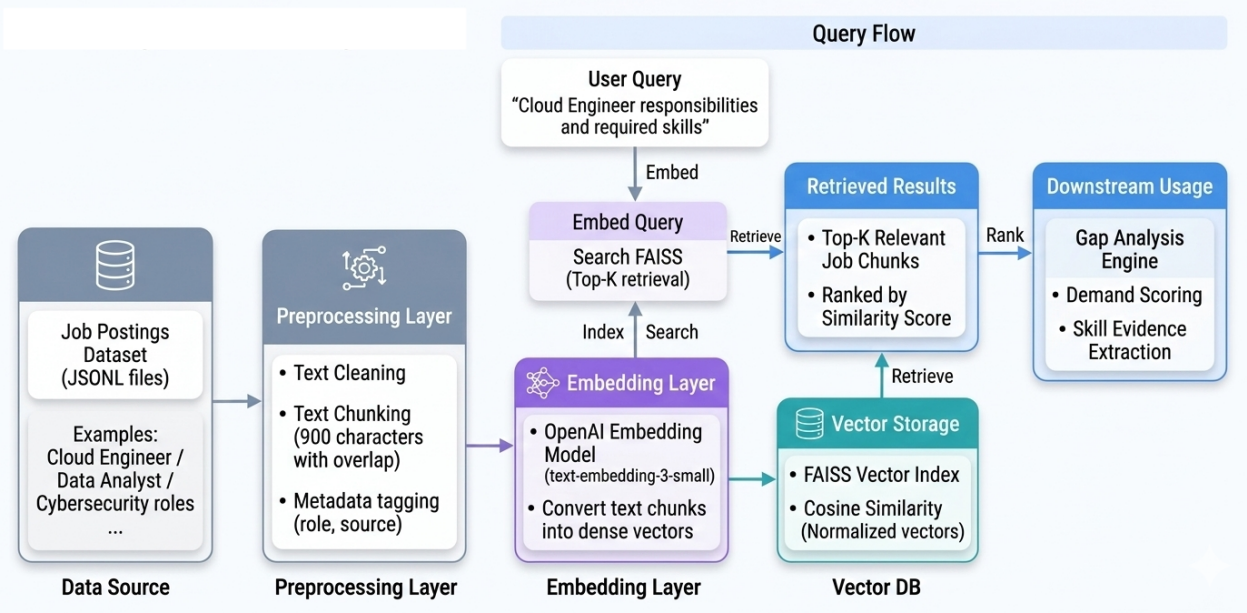

Figure 3: Semantic Job Retrieval Pipeline

2. Embedding generation,
3. FAISS-based vector indexing,
4. Top-K similarity retrieval.

This enables semantic retrieval rather than simple keyword matching, improving relevance of retrieved job requirements.

6 Demand-Weighted Gap Analysis Engine

The gap engine is deterministic and explainable.

6.1 Demand Scoring

For each skill in the taxonomy:

- Count mentions across retrieved job chunks,
- Assign demand score,
- Store supporting evidence snippets.

Skills below a demand threshold are filtered to reduce noise.

6.2 User Evidence Scoring

The system counts skill mentions within the user's resume and GitHub text. Skills are classified as:

- **Strong** (2 mentions),
- **Exposure** (1 mention),
- **Missing** (0 mentions).

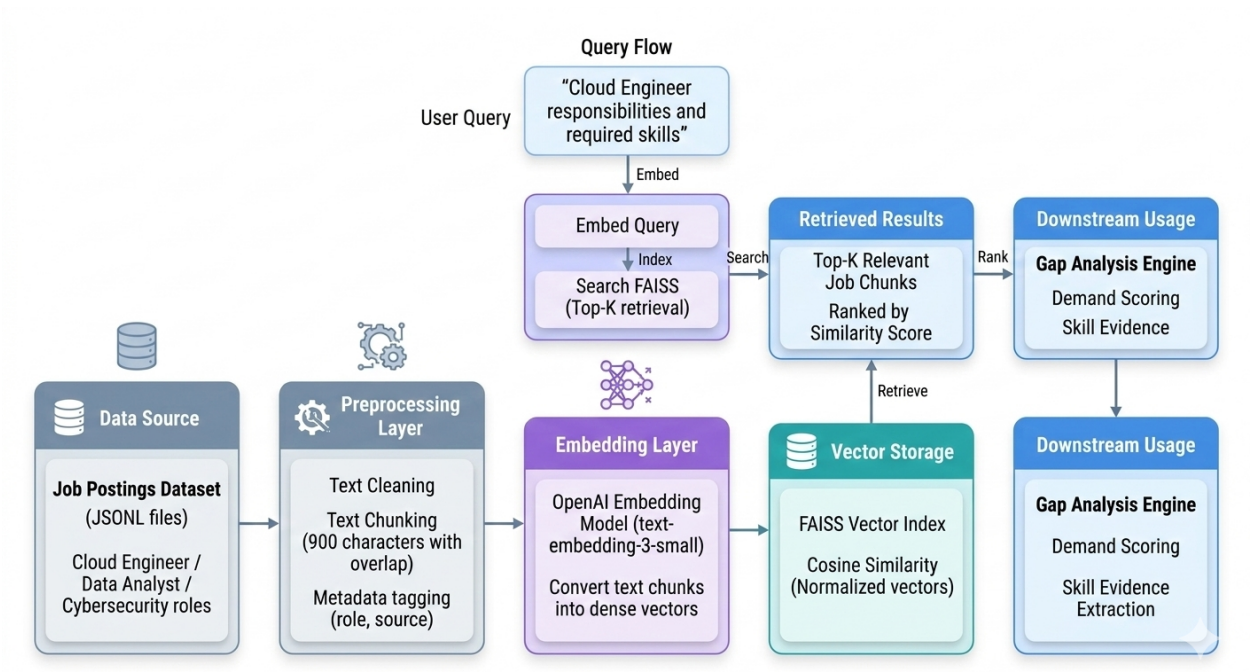

Figure 4: Data Flow and Gap Scoring

6.3 Ranking and Output

Skills are ranked by demand score and grouped into:

- Top Missing Skills,
- Exposure Skills,
- Strengths.

Each missing skill includes evidence from job postings for transparency.

7 Engineering Tradeoffs

Given time constraints, priority was placed on:

- Modular architecture,
- Deterministic scoring,
- AI fallback reliability,
- Input validation and error handling.

The following were intentionally deprioritized:

- Production deployment,
- Real-time job ingestion,
- Advanced ranking heuristics,
- Extensive automated testing.

8 Limitations and Future Work

8.1 Current Limitations

While the system achieves the intended end-to-end functionality, several limitations remain due to scope and time constraints:

- **Static Job Dataset:** Job postings are synthetic and locally stored. The system does not ingest live or continuously updated job data.
- **Heuristic Demand Scoring:** Demand is calculated using frequency-based counting across retrieved chunks. This does not differentiate between senior and junior requirements or weight importance by job title context.
- **Limited Evaluation Metrics:** There is no quantitative benchmarking of retrieval quality or roadmap effectiveness.
- **Basic Testing Coverage:** Only core behavior (happy path and edge case) is tested. Broader integration and stress testing are not implemented.
- **No Persistent User State:** The current system does not store historical user profiles or track skill progression over time.

8.2 What Did Not Work as Intended

During development, several approaches were explored but refined or replaced:

- **Fully LLM-Based Gap Scoring:** An early iteration relied on prompting the LLM to directly identify skill gaps. This approach was rejected due to reduced reproducibility and lack of explainability. Deterministic demand scoring was implemented instead.
- **Direct Resume-to-Job Comparison Without Retrieval:** Comparing resume text directly against the entire job dataset proved inefficient and noisy. Introducing semantic retrieval significantly improved signal quality.
- **Overly Complex Scoring Heuristics:** Attempts to implement weighted scoring based on keyword proximity added complexity without proportional clarity. A simpler frequency-based model was retained for transparency.

8.3 Future Improvements

Given additional development time, the following enhancements would be prioritized:

- **Live Job Ingestion Pipeline:** Implement scheduled scraping or API-based ingestion of job postings with periodic FAISS index rebuilding.
- **Skill Seniority Weighting:** Differentiate entry-level vs. senior-level requirements and weight demand scores accordingly.
- **Advanced Retrieval Tuning:** Introduce evaluation metrics such as precision@k and experiment with hybrid lexical + semantic retrieval.
- **User Progress Tracking:** Store user sessions and track roadmap completion over time.

- **Expanded Testing:** Add integration tests, regression tests, and property-based testing for edge cases.
- **CI/CD Integration:** Add automated linting, testing, and security checks in a continuous integration pipeline.
- **Performance Optimization:** Cache embeddings and optimize index loading for faster startup times.

8.4 Scalability Considerations

To support larger datasets and concurrent users, future architectural adjustments would include:

- Migrating from local FAISS indexing to a managed vector database.
- Introducing asynchronous job retrieval.
- Decoupling the UI from backend services via an API layer.

These improvements would transition the system from a prototype-level implementation to a production-ready platform.

9 Conclusion

Skill Bridge demonstrates a balanced system design that integrates AI capabilities while maintaining deterministic, explainable control over decision-making. The modular architecture allows for extensibility, and fallback mechanisms ensure robustness under degraded conditions.

The system prioritizes clarity, reliability, and engineering discipline over superficial automation, resulting in a reproducible and defensible career intelligence framework.